

Literature Review

University Hall Management System

The management of accommodation of a large number of university students involves a number of activities, including the maintenance of student information, processing of hall applications, allocation of rooms and seats, handling of complaints, and monitoring of administrative activities. In many universities, these activities can become challenging to manage when they are heavily dependent on manual processes. As such, a number of web-based hostel and residential management systems have been developed to facilitate the organization and accessibility of accommodation management.

Wong and Mahdin (2022) developed an Online Hall Management System which included student management, hall applications, room allocation and complaint management. Their study demonstrated how a web-based system could support the core activities of hall administration. The proposed University Hall Management System builds upon similar ideas but incorporates additional functions such as application approval, room and seat management, dashboards and email communication into the same system.

Zamri and Mohd Zaki (2023) developed a Residential Booking System which focused on student information, room availability, booking, and room allotment. Their work highlighted the value of a digital system for managing accommodation and room allocation. Building upon this concept, our system links the hall application process with admin review, approval or rejection, application status tracking, and room allocation.

Dahri and Mohd Yusof (2024) proposed an Online Damage Complaint Management System for university residential facilities. The primary aim of their work was to streamline the reporting and management of accommodation-related complaints. In our system, complaint management is incorporated as one part of the overall hall-management process. This enables complaint information to be handled in conjunction with student, hall, room and administrative information.

Muniandy and Zainal (2025) developed RCARS, a Residential College Accommodation Rental System which covered accommodation applications, application status, room availability, room allocation and reporting. These functions are closely related to the accommodation process of our proposed system. However, our system incorporates them with student and admin dashboards, role-based access, complaint management, room and seat management, and email notification.

Yusof et al. (2026) proposed StaySmart, a web-based hostel management system which covered hostel applications, room assignment, occupancy monitoring, complaints, and role-based access. This study provides a recent example of how different hostel activities can be managed through a web-based platform. Our research follows a similar direction but focuses specifically on university hall administration, incorporating student management, hall applications, approval, room and seat allocation, complaint management, dashboards, tracking, reporting, and email communication.

Research Gap

The reviewed studies cover many important areas of hostel and accommodation management. However, their main focus and combination of features are different. Some systems concentrate mainly on applications and room allocation, while others focus more on complaints or accommodation booking. From this review, the following areas were identified as important for further integration in a university hall management system.

Identified Gap

How Our System Addresses It

1. Different activities are handled separately — Centralized Management: Student, Hall, Room,

Identified Gap	How Our System Addresses It
Existing studies often focus on particular areas of hostel management.	Application, and Complaint Management are handled within the same system.
2. Application processing is not always covered as a complete workflow	Complete Application Process: Students submit applications, administrators review them, and applications can be approved or rejected. Students can also track their application status.
3. Room management and allocation need better integration	Room & Seat Management: The system maintains room and seat information and keeps track of occupied and available seats during allocation.
4. Students and administrators have different requirements	Role-Based Access: Separate access and functions are provided for Admin and Student users.
5. Students need direct access to their accommodation information	Student Dashboard: Students can view their application status, hall information, and allocated room and seat details.
6. Complaint handling can be treated as a separate activity	Complaint & Maintenance Management: Students can submit complaints and administrators can manage them within the same hall-management system.
7. Administrators need a quick overview of hall activities	Admin Dashboard & Reports: Important statistics such as students, rooms, seats, applications, and recent applications can be viewed from the dashboard.
8. Students need timely information about application decisions	Email Notification: Important application updates, including approval or rejection, can be communicated to students through email.

Research Contribution

The main contribution of the proposed **University Hall Management System** is not a single individual feature. Rather, it is the way different hall-management activities are connected within one system.

The overall process can be represented as:

Student Information Management

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Online Hall Application

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Application Review by Admin

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Approval / Rejection

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Application Status Tracking

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Room & Seat Management

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Room & Seat Allocation

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Student Dashboard & Room Details

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Complaint & Maintenance Management

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Admin Dashboard & Statistics



Recent Applications & Reports



Email Notification

Through this approach, students can complete important accommodation-related activities online, while administrators can manage applications, rooms, seats, complaints, and other hall information from a centralized platform.

The proposed system therefore builds on the ideas presented in previous hostel-management studies and combines them into a unified solution for **university hall administration and student accommodation management**.

References

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